

Name:	Elsa [REDACTED]
Instructor:	Samuel Chukwuemeka
Objective:	To convert a measurement from a unit to another unit.
Measurement:	Volume
1st: Given Unit:	Customary unit (Fluid Ounce)
To Convert to:	Metric unit (Milliliters)
2nd: Given Unit:	Metric unit (Milliliters)
To Convert to:	Customary unit (Fluid Ounce)
Container Used:	Softsoap body soap bottle (Please see bottom.)



(1.) **Conversion: 20 fl. oz to mL**

From Given Table: Customary to Customary Conversions

$$1 \text{ qt} = 32 \text{ fl. oz}$$

$$1 \text{ gal} = 4 \text{ qt}$$

From Given Table: Metric to Customary Conversions

$$1 \text{ L} = 0.26417205 \text{ gal}$$

The first conversion used will be from Customary fluid ounces to Metric Liters.

Using the unity fraction method.

$$20 \text{ fl. oz} \times \frac{1 \text{ qt}}{32 \text{ fl. oz}} = 0.625 \text{ qt}$$

$$0.625 \text{ qt} \times \frac{1 \text{ gal}}{4 \text{ qt}} = 0.15625 \text{ gal}$$

$$0.15625 \text{ gal} \times \frac{1 \text{ L}}{0.26417205 \text{ gal}} = 0.5914705965 \text{ L}$$

From Given Table: Metric to Metric Conversion

$$1 \text{ mL} = 10^{-3} \text{ L}$$

Conversion from Liters to Milliliters using the unity fraction method.

$$0.5914705965 \text{ L} \times \frac{1 \text{ mL}}{10^{-3} \text{ L}} = 591.4705965 \text{ mL}$$

Also, equal to the following:

$$0.5914705965 \text{ L} \times \frac{1 \text{ mL}}{0.001 \text{ L}} = 591.4705965 \text{ mL}$$

Rounding to the nearest whole number:

$$\approx 591 \text{ mL}$$

Confirming the quantity of fluid ounces to milliliters from the body soft-soap container.

(2.) Conversion: 591 mL to fl.oz

From Given Table: Metric to Metric Conversion

$$1 \text{ mL} = 10^{-3} \text{ L}$$

First conversion from Milliliters to Liters using the unity fraction method.

$$591 \text{ mL} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} = 0.591 \text{ L}$$

From Given Table: Metric to Customary Conversions

$$1 \text{ L} = 0.26417205 \text{ gal}$$

From Given Table: Customary to Customary Conversions

$$1 \text{ gal} = 4 \text{ qt}$$

$$1 \text{ qt} = 32 \text{ fl. oz}$$

The second conversion used will be from Metric Liters to Customary fluid ounces.

Using the unity fraction method.

$$0.591 \text{ L} \times \frac{0.26417205 \text{ gal}}{1 \text{ L}} = 0.1561256816 \text{ gal}$$

$$0.1561256816 \text{ gal} \times \frac{4 \text{ qt}}{1 \text{ gal}} = 0.6245027262 \text{ qt}$$

$$0.6245027262 \text{ qt} \times \frac{32 \text{ fl. oz}}{1 \text{ qt}} = 19.98408724 \text{ fl. oz}$$

Rounding to the nearest whole number:

$$\approx 20 \text{ fl. oz}$$

Confirming the quantity of milliliters to fluid ounces from the body soft-soap container.